# CAPSTONE PROJECT

## Music Playlist & Listening Stats Manager


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**SECTION - 2**




# Abstract



The **Music Playlist and Listening-Stats Manager** is a simple Java-based application used to manage songs and track listening habits. It allows users to add songs, store song details, display songs, and record the number of times each song is played. It provides an easy way to organise music and understand which songs are played the most. In the future, features like **song search, favourites, sorting, multiple playlists, and a graphical interface** can be added.

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# Introduction

## What It Is

A simple Java-based application to manage songs and track listening habits with ease.

## What It Does

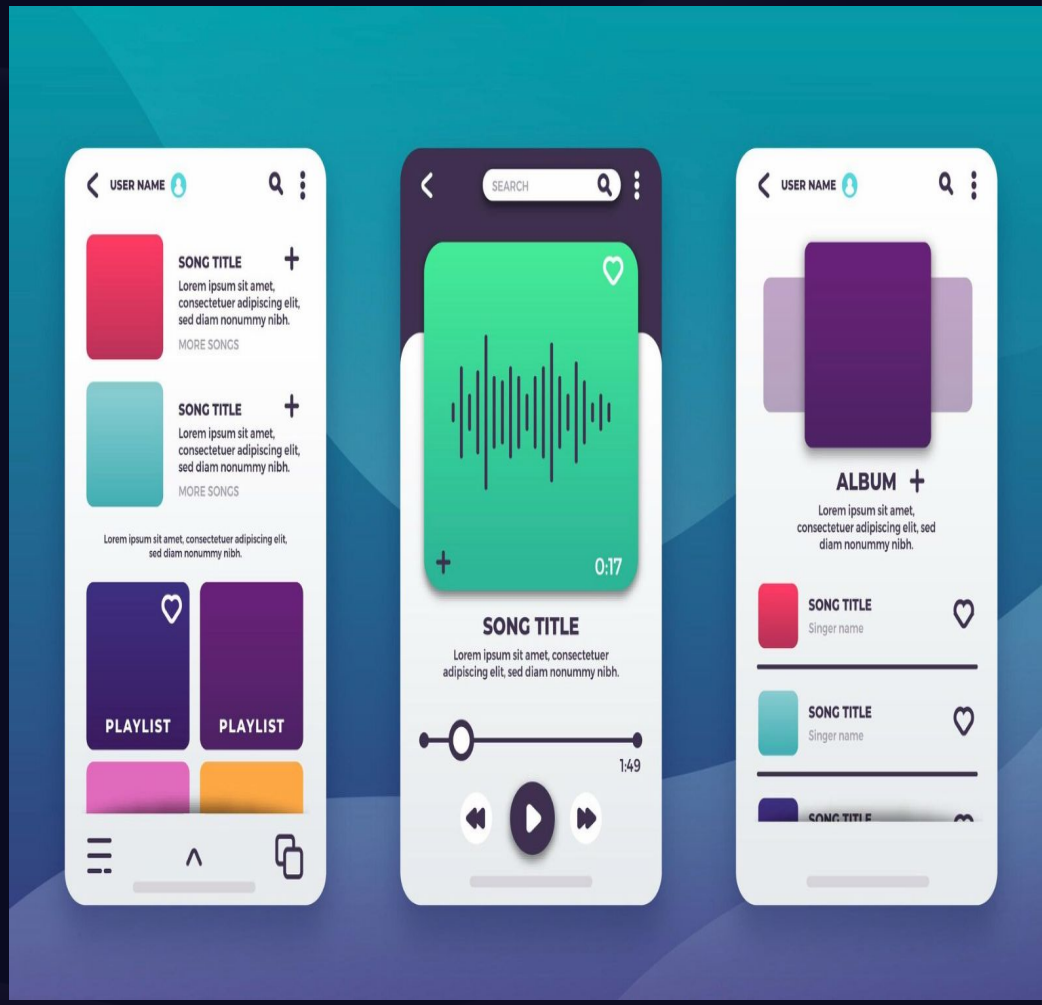
Add songs, store song details, display your music collection, and count plays per song.

## How It Works

Built using arrays, loops, conditions, switch statements, operators, and methods.

## Where It Goes

Future scope includes song search, sorting, multiple playlists, favourites, and a GUI.




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# Course Outcomes Covered

### CO1 — Data Types & Operators

Data types and operators are used to store song details such as name, duration, and play count, and to perform calculations on listening statistics.

### CO2 — Control Structures

Conditions, loops, switch statements, and break are used to control program flow, handle user choices, and manage the interactive menu system.

### CO3 — Methods & Arrays

Methods organize different operations such as adding and displaying songs, while arrays efficiently store and manage multiple song details in memory.

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# Course Outcomes Covered

## CO4 — OOP & Exceptions

Song and Playlist classes with encapsulated fields, inheritance for song types, and try-catch for invalid input.

## CO5— Strings & File I/O

String methods to search by title or artist, Comparator to sort by play count, and files to save playlists.

## CO6 — Collections & Streams

ArrayList and HashMap to store songs and play counts, with Streams to filter, sort, and group listening stats.

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# Project Objectives




## Primary Goal

Develop a **user-friendly platform** for playlist management and music analytics that is accessible, organised, and insightful.

## Core Focus Areas

- Playlist creation and custom organization
- Real-time listening statistics tracking
- Music consumption pattern analysis
- Secure authentication and data persistence
- Music search and discovery functionality

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# Advantages

## User-Centric Design

Intuitive interface makes playlist management effortless for all users.

## Real-Time Analytics

Instant insights into listening habits, preferences, and play counts.

## Data Visualization

Charts and graphs make statistics easy to interpret at a glance.

## Security

Secure authentication protects user data and privacy effectively.

## Accessibility

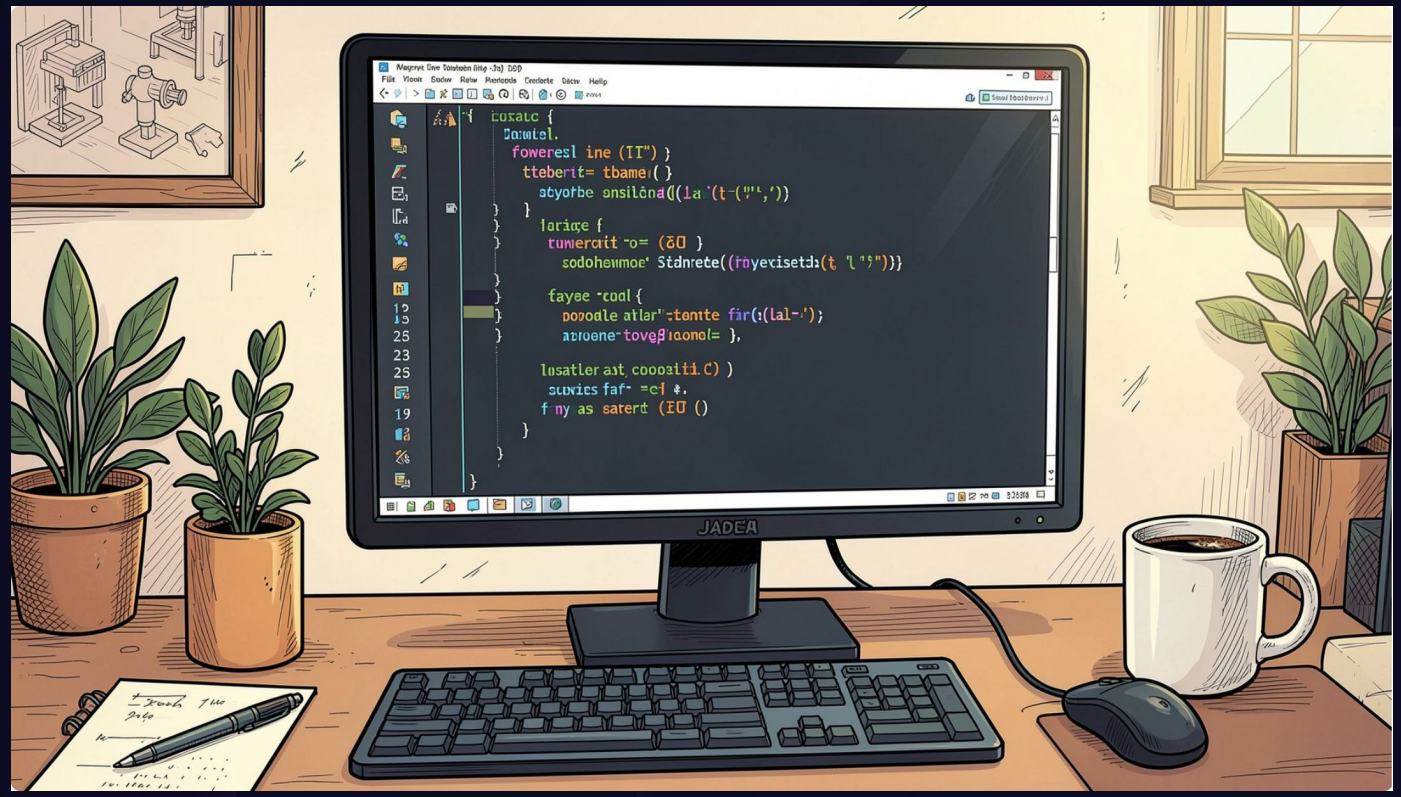
Web-based platform accessible across all devices and browsers.

## Personalization

Customizable playlists and preference tracking for every user.

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# Project Development



## Built Entirely in Java

The project was developed using core Java concepts taught during the course, without any external libraries or frameworks

1 **Arrays**
Used to store song names, durations, and play counts.

2 **Methods**
Separate methods handle adding, displaying, and updating songs.

3 **Loops & Conditions**
Control program flow and handle user input validation.

4 **Switch Statement**
Powers the interactive menu for navigating features.

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# Implementation & Demo

A walkthrough of the core features built into the Music Playlist & Listening Stats Manager:

01

## User Registration & Login

Secure user registration and login functionality to protect personal data.

02

## Playlist Management

Create, edit, and delete playlists — and add or remove songs with ease.

03

## Listening Statistics Dashboard

Real-time dashboard tracking play counts, top songs, and listening trends.

04

## Search & Filter

Search and filter music by genre, artist, or title for quick discovery.

05

## CRUD Operations

Full Create, Read, Update, and Delete operations demonstrated on playlists and user data.

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# CONCLUSION




To conclude, our Music Playlist and Listening Stats Manager provides a simple and organised way to manage songs and playlists.

It helps users keep track of their listening activity and understand their music preferences. Through this project, we also applied important Java concepts such as arrays, loops, conditions, switch statements, and methods to a real-world application.
In the future, we can improve the project by adding features like song search, sorting, multiple playlists, favourites and a graphical user interface.

**Overall, this project helped us improve our Java programming skills and understand how programming can be used to create useful applications**

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# Future Scope

1

### AI Recommendations

Machine learning to suggest songs based on listening patterns.

2

### Social Features

Share playlists, collaborate with friends, discover trending music.

3

### Mobile App

Native iOS and Android apps for enhanced mobile experience.

4

### Streaming Integration

Direct integration with Spotify, Apple Music, and YouTube Music.

5

### Offline Mode & Gamification

Download playlists and earn achievements, badges, and leaderboards.

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# Thank you



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